

Mohammad Niroomand

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RESEARCH INTERESTS

- Wireless and Mobile Networks
- 5G & 6G Networks
- Multimedia Networks
- Network Security
- Reinforcement Learning

EDUCATION

K. N. Toosi University of Technology

Sep 2024 – Current

M.Sc. in Computer Engineering, Minor in Computer Networks

Tehran, Iran

Courses: Advanced Computer Networks, Wireless Networks, Multimedia Networks, Performance Evaluation of Computer Systems, Network Management, Advanced Network Security, Cloud Computing, Distributed Systems

Seminar Title: The Role of Non-Terrestrial Networks (NTN) in the Expansion of 5G and Beyond.

Master's Thesis (Supervisor: [Prof. Kaebbeh Yaeghoobi](#)):

- Thesis Title: Performance optimization between terrestrial 5G and LEO satellite networks using reinforcement learning.

University of Guilan

Sep 2019 – Jul 2024

B.Sc. in Computer Engineering

Rasht, Iran

Notable Courses: Computer Networks (4/4), Embedded and Real-Time Systems (4/4), Special Topics 1 (Internet of Things) (4/4), Internet Engineering (4/4), Computer Architecture (4/4), Algorithms Design (4/4), Signal and Systems (4/4)

Bachelor's Project (Supervisor: [Prof. Mohammad Salehi](#)):

- Project summary: Development of a responsive web based weather client application, and leveraging AccuWeather API for real-time Weather data retrieval, forecast visualization and user centric interactive features, together with their design.
- GPA: 4/4

Tadbir High School

Sep 2018 – Sep 2019

High School Diploma in Mathematics & Physics

Qazvin, Iran

- Major GPA: 4/4 (19.10/20)

HONORS AND AWARDS

- Accepted into the Master's program at K. N. Toosi University of Technology in Tehran, **one of the top three institutes of technology** for Computer Networks and Telecommunication Systems in Iran.
- **Ranked in the top 3 %** among more than 16,000 participants in the Iranian National University Entrance Exam (Konkour) for M.Sc. of Computer Engineering, February 2024.
- **Ranked in the top 4 %** among more than 150,000 participants in the Iranian National University Entrance Exam (Konkour) for B.Sc. of Computer Engineering, July 2019.
- Received competitive job offers as **Network Specialist at the Central Bank of Iran** and **Infrastructure Engineer at MTN Irancell**.

- **National LEGO Robotics Team Member** & Qualifier for the 2015 World Robot Olympiad (WRO).
- **Full tuition waiver scholarship** - University of Guilan, 2019-2024

ACADEMIC PROJECTS

- **Router Queue Performance Analysis** [\[Link\]](#)
Simulated and analytically evaluated router queuing systems to analyze packet delay and loss using queuing theory.
- **Advanced Network Security Lab** [\[Link\]](#)
The five project components were implemented and simulated, including: OSINT & phishing simulation, Nmap scanning, Snort IDS analysis, DoS testing, reverse shell with traffic obfuscation, and HTTPS/TLS certificate & cipher-suite assessment.
- **Hardware Software Co-Design** [\[Link\]](#)
Design and Implementation of a 6-bit Processor with Programming Capability.
- **Embedded and Real-Time Systems** [\[Link\]](#)
Design and simulate an air conditioning system using Proteus. Measure temperature and humidity from three different points using sensors and control a heater, a cooler, and a humidifier accordingly.
- **Microprocessors and Assembly Language** [\[Link\]](#)
Design a traffic light control system at an intersection using microprocessors and assembly language. Simulate the lights in Proteus with LEDs and implement a 7-segment countdown timer.
- **Sentiment Analysis** [\[Link\]](#)
Build both traditional machine learning models (Naïve Bayes/Logistic Regression) and a TensorFlow neural network for sentiment analysis on the Amazon reviews dataset.

TEACHING EXPERIENCE

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|---|--|
| ▪ Teaching Assistant at K. N. Toosi University of Technology | Sep 2025 - Currently |
| - Network Management (Spring 2026) | <i>Supervisor: Prof. Kaebbeh Yaeghoobi</i> |
| - Computer Networks I (Spring 2026) | <i>Supervisor: Prof. Kaebbeh Yaeghoobi</i> |
| - Digital Electronics (Spring 2026) | <i>Supervisor: Prof. Zeinab Seifoori</i> |
| - Technical English (Spring 2026) | <i>Supervisor: Prof. Kaebbeh Yaeghoobi</i> |
| - Advanced Computer Networks (Fall 2025) | <i>Supervisor: Prof. Kaebbeh Yaeghoobi</i> |
| ▪ Teaching Assistant at University of Guilan | Sep 2019 - Jul 2024 |
| - Embedded and Real-Time Systems (Spring 2024) | <i>Supervisor: Prof. Mohammad Salehi</i> |
| - Algorithm Design (Spring 2023) | <i>Supervisor: Prof. Aida Khozaei</i> |
| - Advanced Programming (Spring 2022) | <i>Supervisor: Prof. Aida Khozaei</i> |

COURSES

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| ▪ Sample-based Learning Methods Verification
<i>University of Alberta</i> | May 2026 |
| ▪ Fundamentals of Reinforcement Learning Verification
<i>University of Alberta</i> | Dec 2025 |
| ▪ CISSP Certified Information Systems Security Professional Verification
<i>Packt</i> | Jul 2025 |
| ▪ Supervised Machine Learning: Regression and Classification Verification
<i>DeepLearning.AI</i> | Dec 2024 |
| ▪ CCNA 200-301 Verification | Nov 2024 |

- **CEH (Certified Ethical Hacker)** | [Verification](#)
Maktabkhooneh

Apr 2024

- **LEGO Education and Advanced Robotics Training** | [Verification](#)
LEGO Education Institute

2014-2015

SKILLS

- **Programming Languages:** Python, Java, C/C++, VHDL/Verilog, Solidity
- **Developer Tools:** EVE-NG, GNS3, Cisco Packet Tracer, Paessler PRTG, Wireshark, Proteus, VS Code, Linux, PyCharm, IntelliJ, Git, Github, Slack, $\LaTeX$
- **Web Development:** React, Bootstrap, HTML/CSS, Tailwind CSS, JavaScript, MySQL
- **English:** IELTS: To be taken soon

PERSONAL INTERESTS

- **Music:** Professional piano player with two years of teaching experience.
- **Literature & Arts:** Interested in poetry and literature, with experience in film, theater, and book criticism. Furthermore, I organized an open studio in Cité Internationale des Arts in Paris (2015)
- **Social & Environmental Engagement:** Active collaboration with charitable and environmental institutions.